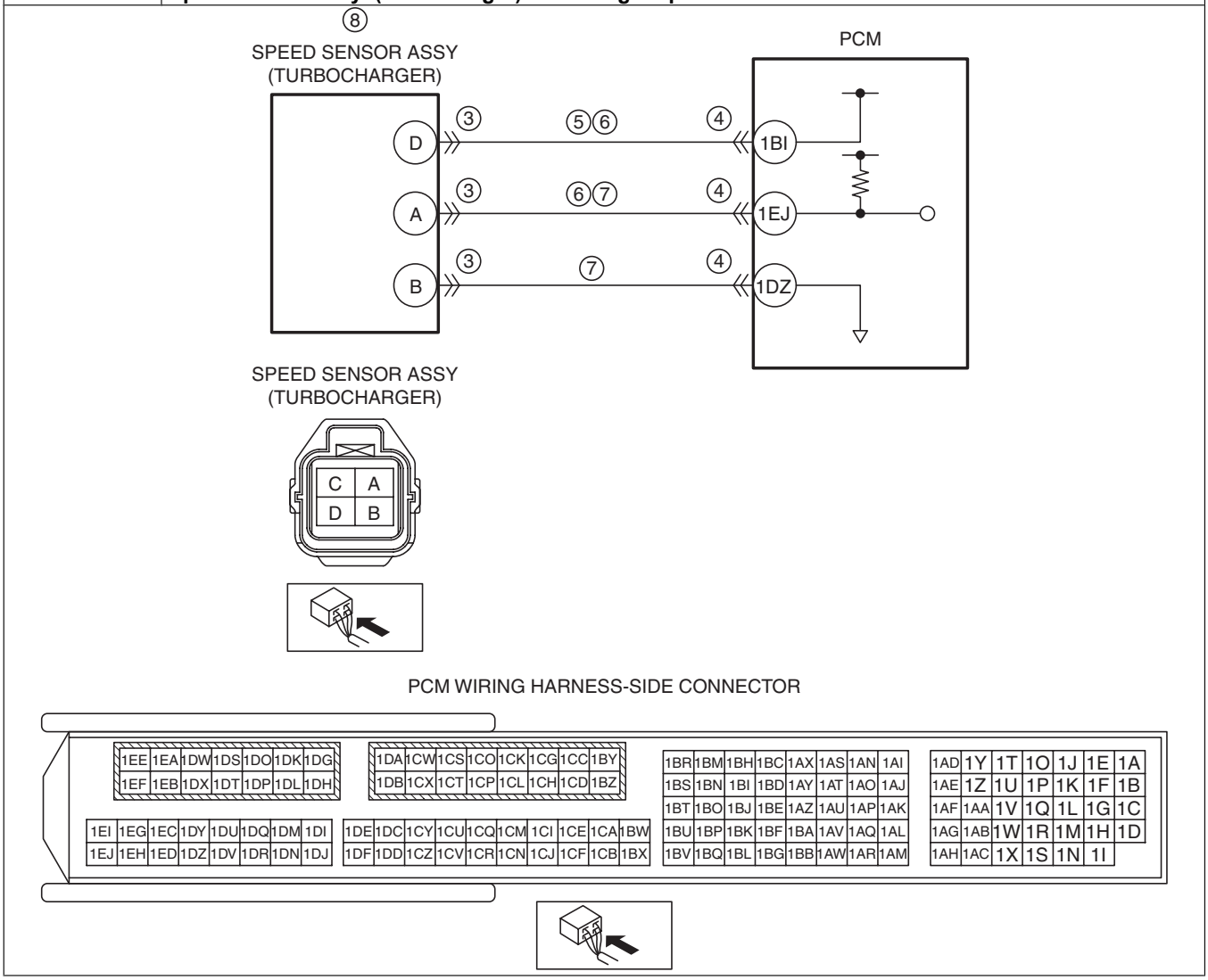
DTC P2581:00 [SKYACTIV-D 1.5]

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DTC P2581:00	Speed sensor assy. (turbocharger) circuit high input
DETECTION CONDITION	• If the input voltage at the PCM terminal 1EJ is **more than 2.5 V** for **3.5 s**, the PCM determines that the speed sensor assy. (turbocharger) circuit is high. **MONITORING CONDITIONS** — **0.5 s** have elapsed after the engine was started. **Diagnostic support note** • This is a continuous monitor (Turbo system). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• Inhibits the EGR control. • Inhibits the auto diesel particulate filter regeneration control. • Inhibits engine-stop by operating the i-stop function.
POSSIBLE CAUSE	• Speed sensor assy. (turbocharger) connector or terminals malfunction • PCM connector or terminals malfunction • Short to power supply in wiring harness between the following terminals: — Speed sensor assy. (turbocharger) terminal A—PCM terminal 1EJ — Speed sensor assy. (turbocharger) terminal B—PCM terminal 1DZ • PCM connector or terminals malfunction • Speed sensor assy. (turbocharger) power supply circuit and signal circuit are shorted to each other • Open circuit in wiring harness between the following terminals: — Speed sensor assy. (turbocharger) terminal A—PCM terminal 1EJ — Speed sensor assy. (turbocharger) terminal B—PCM terminal 1DZ • Speed sensor assy. (turbocharger) malfunction • PCM malfunction

DTC P2581:00	Speed sensor assy. (turbocharger) circuit high input



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS WHEN DTC WAS DETECTED TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture of the PC function. • Record the freeze frame data/snap shot data.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT SPEED SENSOR ASSY. (TURBOCHARGER) CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the speed sensor assy. (turbocharger) connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.

STEP	INSPECTION		ACTION
4	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.
5	**INSPECT SPEED SENSOR ASSY. (TURBOCHARGER)** • Reconnect all disconnected connectors. • Inspect the speed sensor assy. (turbocharger). (See SPEED SENSOR ASSY (TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY) INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the speed sensor assy. (turbocharger), then go to the next step. (See SPEED SENSOR ASSY (TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY) REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
6	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Perform the KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
7	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.